

2g. Content of Additional Pure Mathematics (Optional paper Y545)

Introduction to Additional Pure Mathematics.

In **Additional Pure Mathematics** learners will broaden and deepen their knowledge of pure mathematics, studying both discrete and continuous topics which form the foundation of undergraduate study in mathematics and mathematical disciplines. This area covers recurrence relations, number theory, group theory, the vector product, surfaces and partial differentiation.

8.06 Further Calculus

The calculus techniques in the Pure Core are extended to finding arc-lengths and surface areas of revolution and the use of reduction formulae.

Assumed knowledge

Learners are assumed to know the content of GCSE (9–1) Mathematics and A Level Mathematics. They are also assumed to know the content of Pure Core (Y540 and Y541). All of this content is assumed, but will only be explicitly assessed where it appears in